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***B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022***

CS/EC/IT 19-200-0102B & CE/EE/ME/SE 19-200-0202A

ENGINEERING PHYSICS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) What is the need of a resonator in a laser device?
- (b) Explain the importance of acceptance angle in fibre communication.
- (c) Define coordination number and packing factor in crystallography.
- (d) Write any three applications of SMA.
- (e) How optical and chemical properties of nanomaterials vary with their size?
- (f) Explain the term cooper pair used in BCS theory of superconductivity.
- (g) Write a note on dual nature of matter and radiation.
- (h) Write any three properties of ultrasonic waves.

PART B

(4 × 12 = 48)

- II. (a) What is population inversion? How it is achieved? (3)
- (b) With the help of suitable diagrams, explain the principle, construction and working of a He-Ne laser? (7)
- (c) A semiconductor diode laser has a peak emission wavelength of 1.55 micrometer. Find its band gap in eV. (2)

OR

- III. (a) Explain the terms Numerical aperture and Acceptance angle. (3)
- (b) With suitable figures, describe graded index optical fibre and explain the transmission of signal through it? (7)
- (c) The numerical aperture of an optical fibre is 0.35. If the difference in the refractive indices of the material of its core and cladding is 0.05, calculate the refractive index of material of the core. (2)

- IV. (a) What are Miller indices? How they are calculated? (3)
- (b) Describe Bragg's X-ray spectrometer and explain how Bragg's law can be verified. (7)
- (c) Calculate the interplanar spacing for (3 2 1) planes in a simple cubic crystal whose lattice constant is 4.2 AU. (2)

OR

- V. (a) Write any two structural and magnetic properties of metallic glasses. (3)
- (b) Mention any two characteristics of liquid crystal materials. Explain the construction and working of liquid crystal displays. (7)
- (c) Sketch the following planes of a cubic unit cell (001), (111). (2)

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- VI. (a) Why nanomaterials exhibit different properties? (3)
 (b) Discuss any two electrical, magnetic and mechanical properties of nanomaterials. (7)
 (c) Mention any two applications of nanotechnology. (2)

OR

- VII. (a) What is the penetration depth of a magnetic field on a superconductor? (3)
 (b) With suitable diagrams, describe the difference between type I and type II superconductors? (7)
 (c) The penetration depth of mercury at 4 K is about $800 \text{ \AA}$. Estimate the penetration depth at 0 K (2)

- VIII. (a) State and explain Heisenberg's uncertainty principle. (3)
 (b) Derive time independent Schrodinger wave equation. (7)
 (c) Calculate the number of photons emitted in 5 hours by a 30 W sodium lamp ($\lambda = 5893 \text{ \AA}$) (2)

OR

- IX. (a) Mention any three properties of ultrasonic waves. (3)
 (b) With suitable diagrams, explain how ultrasonic waves can be produced using piezoelectric crystal. (7)
 (c) Calculate the fundamental frequency of a 4mm thick quartz crystal. Young's modulus for quartz $7.9 \times 10^{10} \text{ N/m}^2$ and density for quartz 2650 Kg m^{-3} . (2)
